

Peripheral Interface Device 8155 (I/O Interface & Timer)

Outline

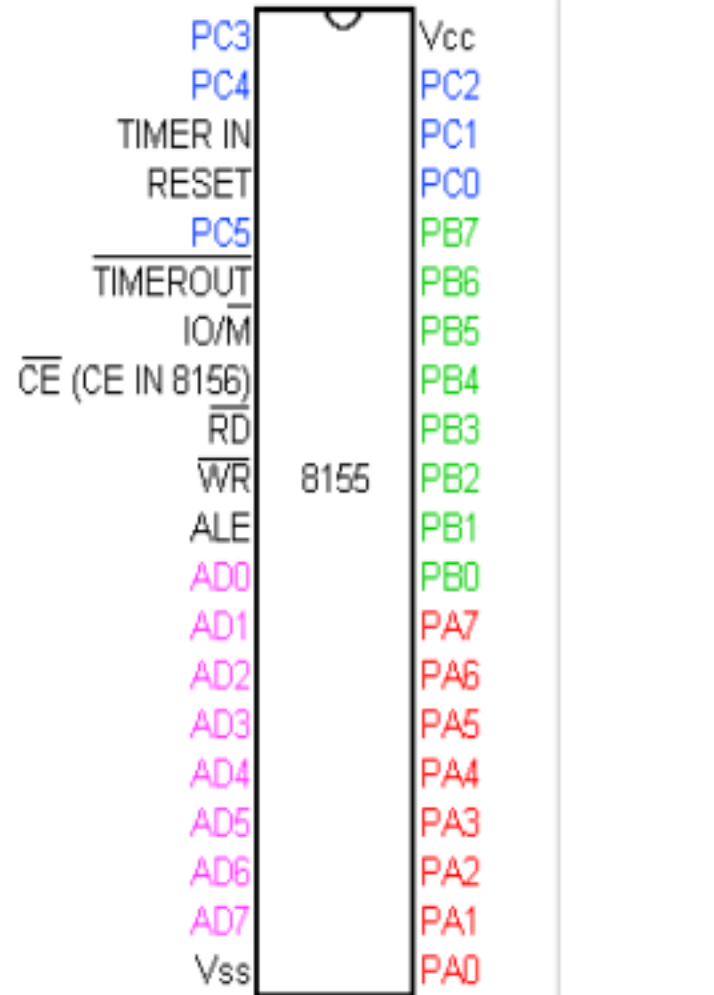
- Review
 - Programmable Interface device 8155
 - Block diagram, Address Calculation diagram
 - Interfacing LED using 8155
- 8155 Timer
 - Modes of timer
 - Square wave generation using 8155 interfaced

timer

- 8155 Handshake & Interrupt mode •
- Interfacing A/D Converter using Handshake mode using 8155

8155 Block Diagram

Reset in



8155

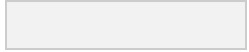
RD
WR
CE
IO/M

AD0-AD7



ALE

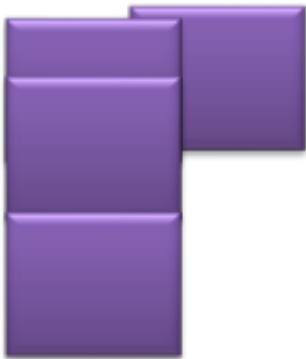
Timer



CLK



RAM



PA0-PA7

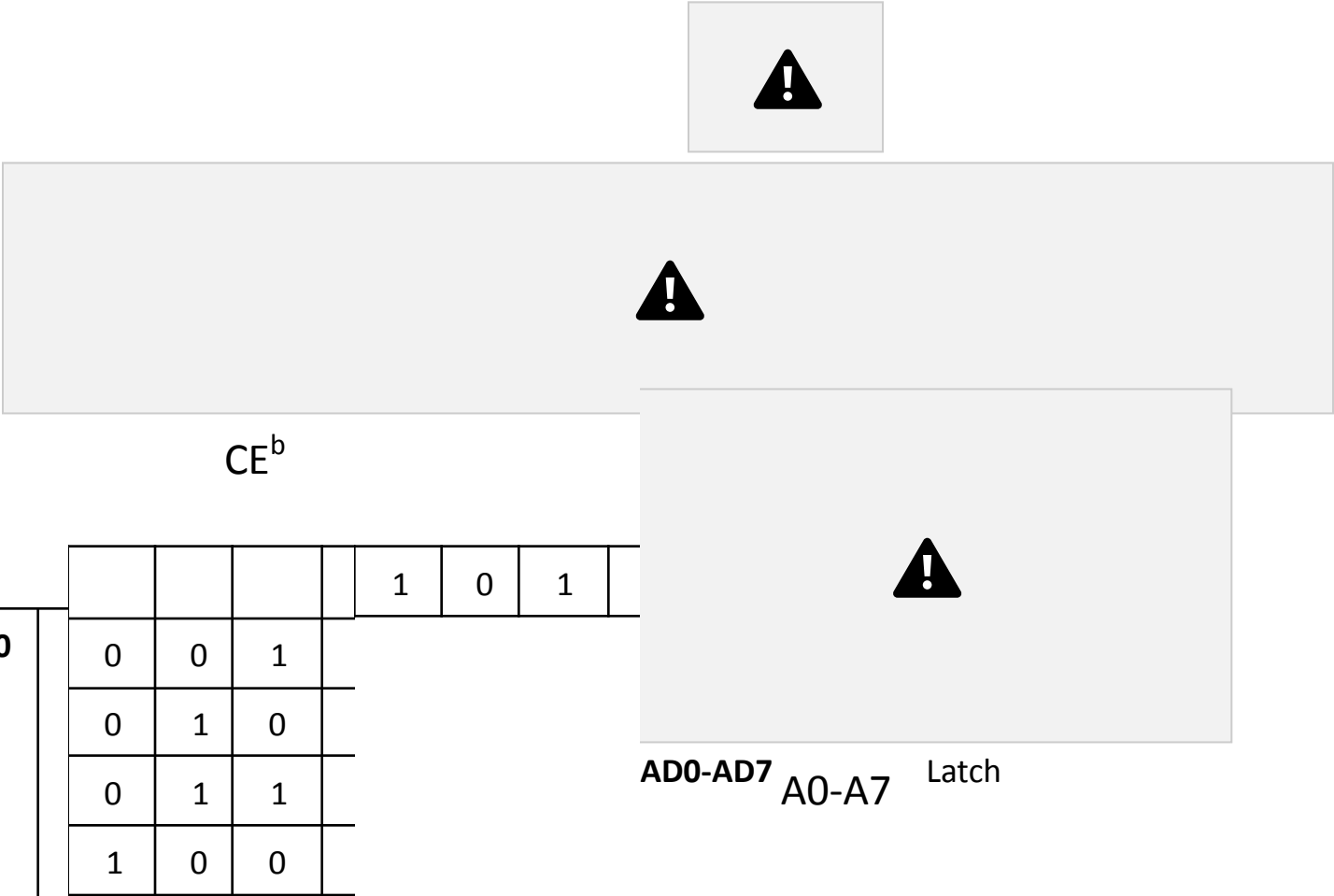


PB0-PB7

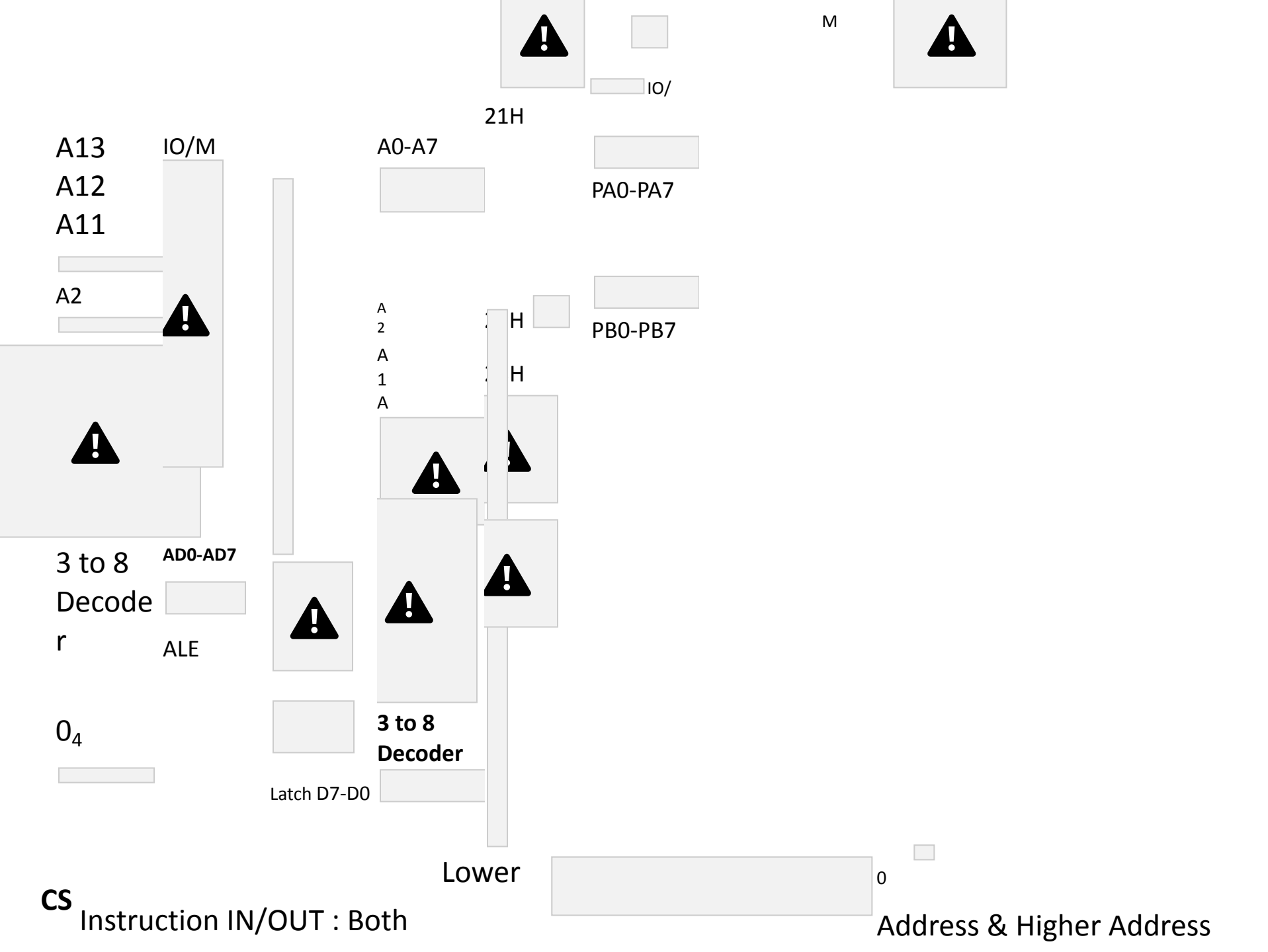


PC0-PC5

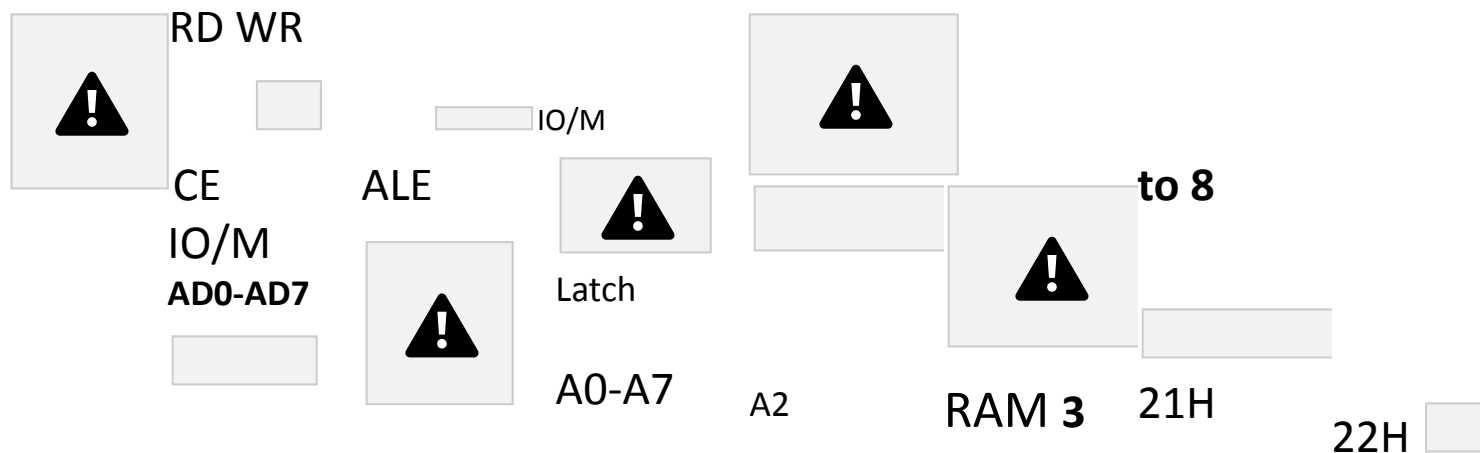
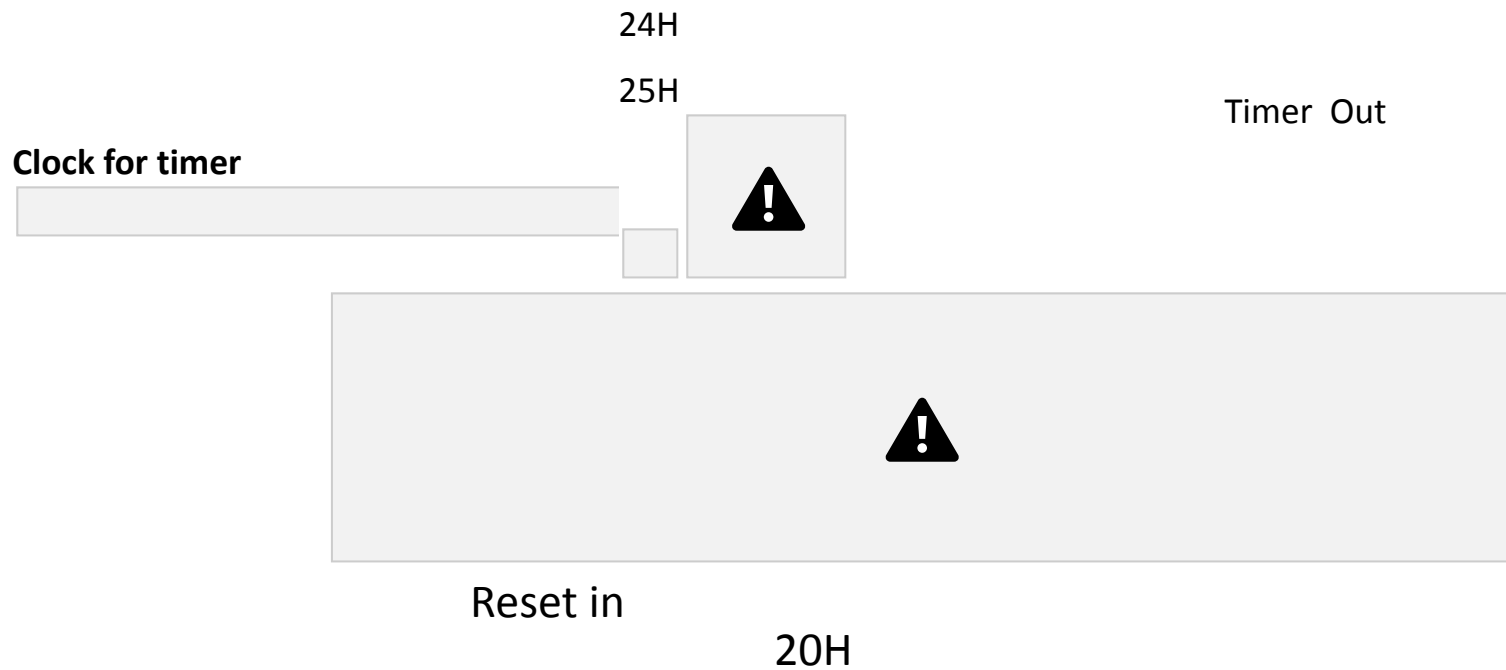
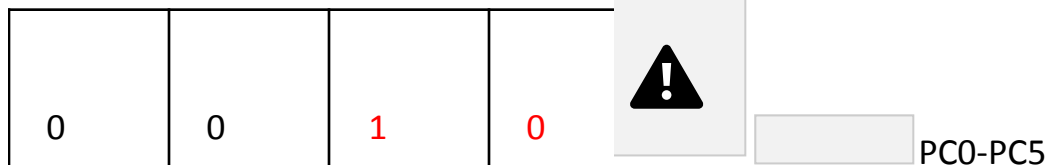
Timer
Out

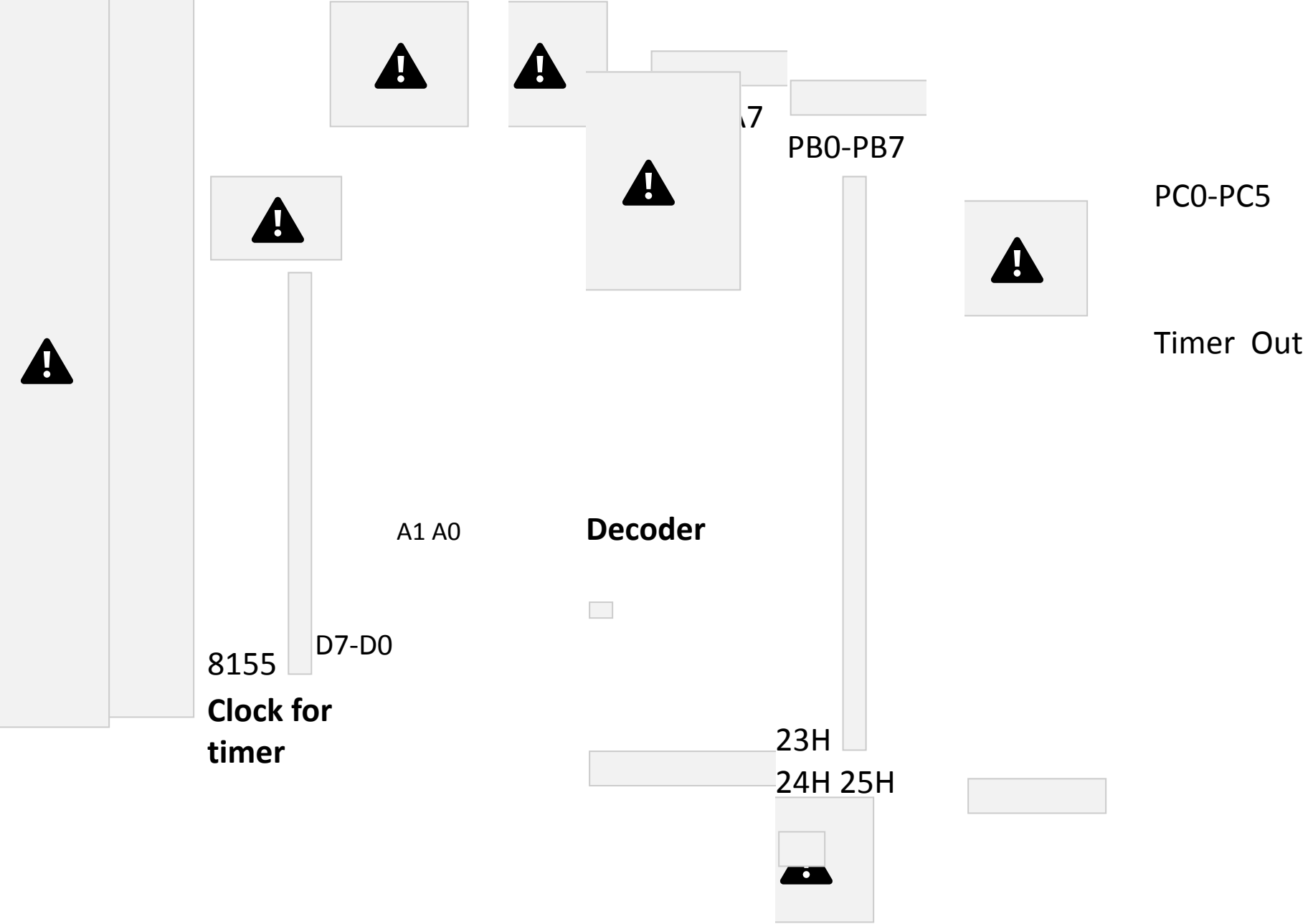






are
same







D7 D6	D5	D4	D3 D2	D1	D0
Timer Command	IEB	IEA	PC	PB	PA

- D0, D1: mode for PA and PB, 0=IN, 1=OUT
- D2, D3: mode for PC
- D4, D5: interrupt EN for PA and PB, 0=disable 1=enable
- D6, D7: Timer command:
 - 00: No effect
 - 01: Stop if running else no effect
 - 10: Stop after terminal count (TC) if running, else no effect
 - 11: Start if not running, reload at TC if running.
- Port C bits (D2, D3)

ALT	D3	D2	PC5
1	0	0	IN
2	0	1	OUT
3	1	0	OUT

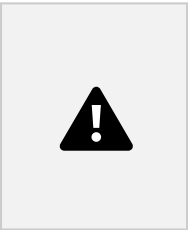
4	1	1	STB _B	
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- Registers

A2	A1	A0	Port (ALE high, AD0=A0)
0	0	0	Command/Status Register
0	0	1	PA
0	1	0	PB
0	1	1	PC
1	0	0	Timer LSB
1	0	1	Timer MSB

A15 A14

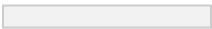


Seg

A13 A12 A11



3 to 8
Decoder



LED



Driver

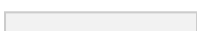


A2

AD7

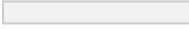
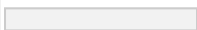


A1 A0



to

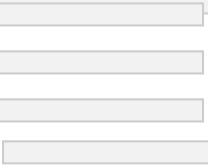
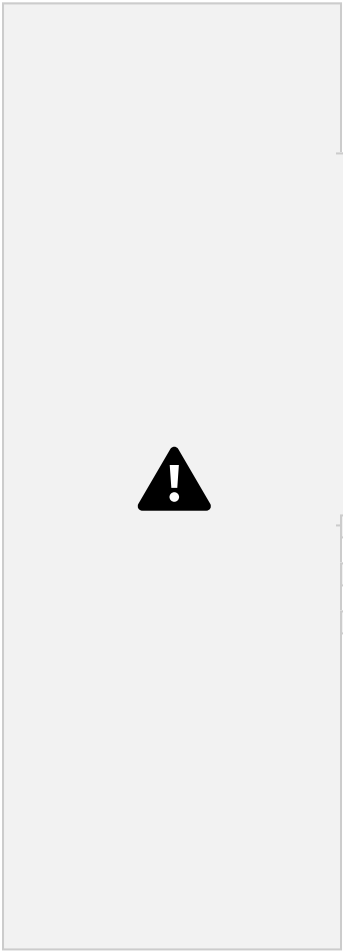
AD0



5V



0₄

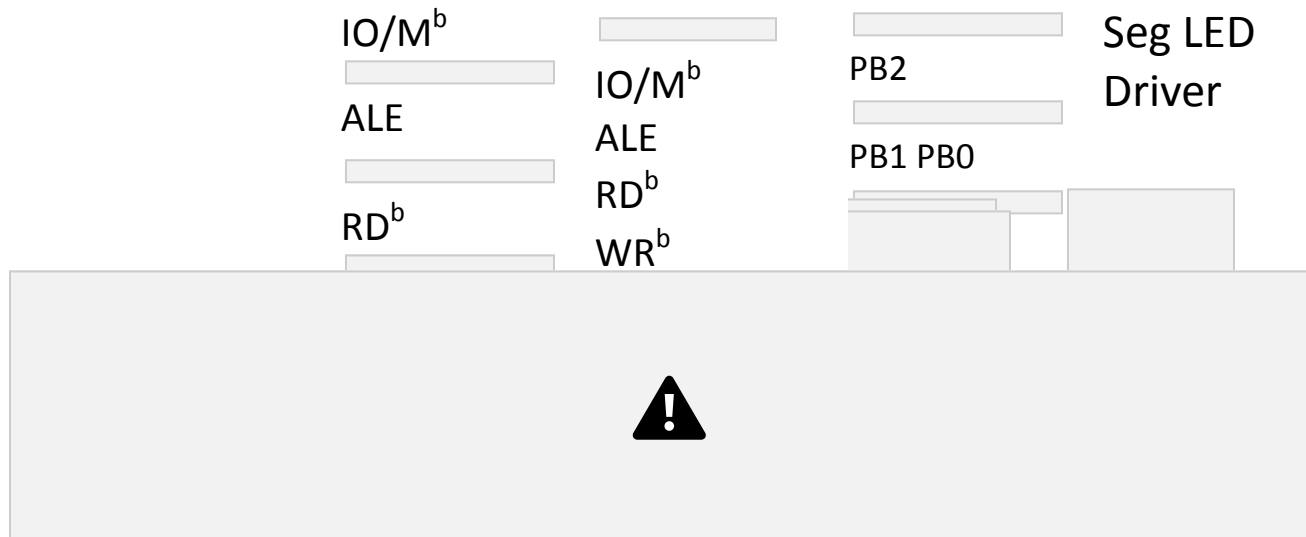


7 Seg
LED
Driver



LED
Driver

7 Seg



- Port Address
 - Control Register=20H, Port A= 21H, Port B= 22H

- Control word:

D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	0	0	0	1	1
Timer		Not Applicable		Use for Port C		Port B Output	Port A Output

- Program

- MVI A,03 ; initialize Port A &B for O/P – OUT 20H
- MVI A, BYTE1 ; Display BYTE1 at port A – OUT 21H
- MVI A, BYTE2 ; Display BYTE2 at port B – OUT 22H



MSB

M2	M1	T13	T12	T11	T10	T9	T8
LSB							
T7	T6	T5	T4	T3	T2	T1	T0

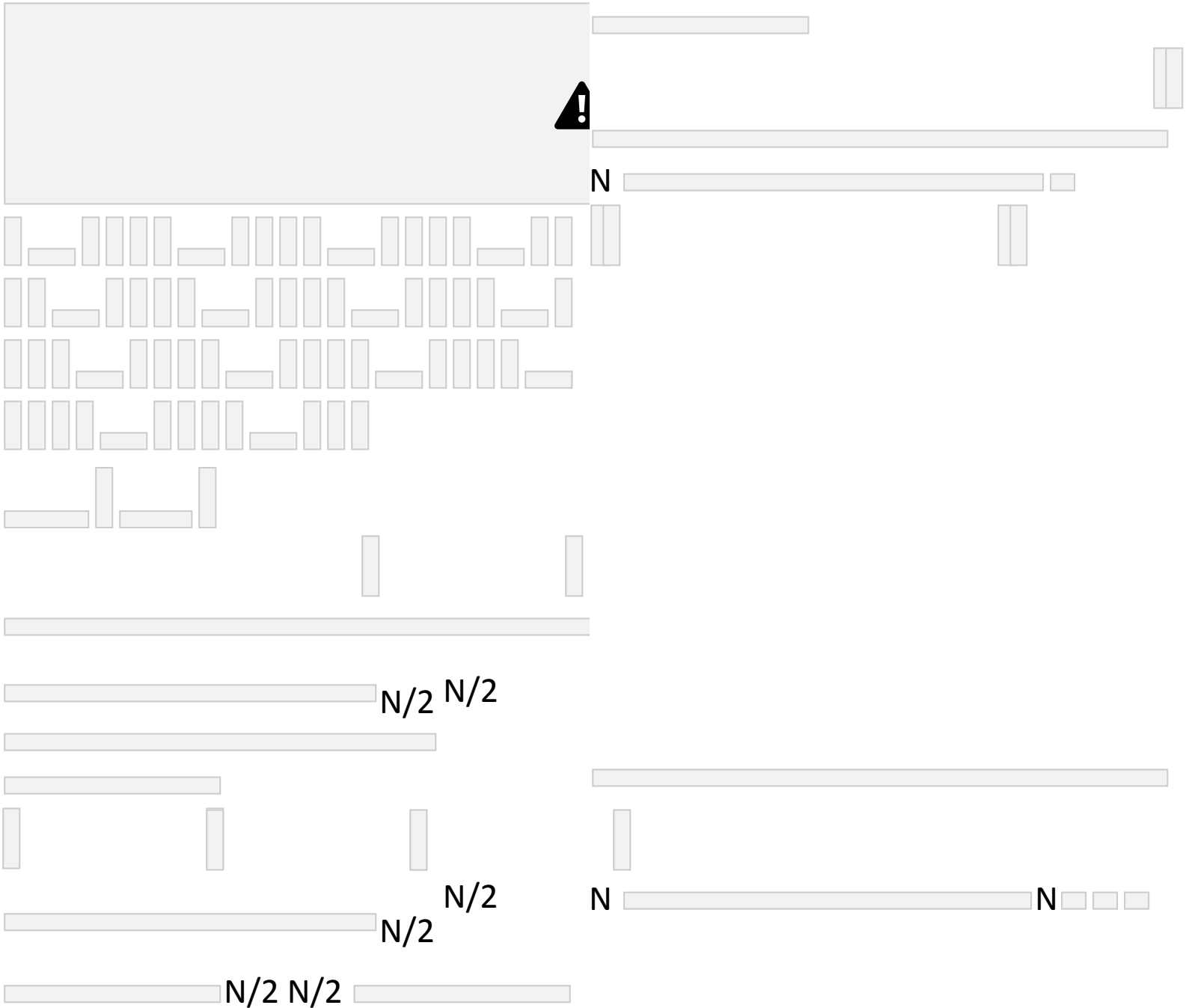
- M2, M1: mode bits:
 - 00: Single square wave of wavelength $TC/2$ ($TC/2, TC/2$ if TC even; $[TC+1/2], [TC-1/2]$ if TC odd)
 - 01: Square waves of wavelength TC ($TC/2, TC/2$ if TC even; $[TC+1/2], [TC-1/2]$ if TC odd)
 - 10: Single pulse *on* the TC 'th clock pulse
 - 11: Single pulse *on* every TC 'th clock pulse.

Mode 00 Mode 01 Mode 10 Mode 11



CLK

WR



- 00: Single square wave of wavelength $TC/2$ ($TC/2, TC/2$ if TC even; $[TC+1/2], [TC-1/2]$ if TC odd) –
- 01: Square waves of wavelength TC ($TC/2, TC/2$ if TC even; $[TC+1/2], [TC-1/2]$ if TC odd)
- 10: Single pulse *on* the TC 'th clock pulse –
- 11: Single pulse *on* every TC 'th clock pulse.



- Design a square wave with pulse width $100\mu S$ •
- Mode 1
- Clock Frequency 3 MHz

Timer count: Pulse Period/Clock period

$$= 200 \times 10^{-6} / 330 \times 10^{-9} = 606 = 25E \text{ H}$$

$$= 02 \text{ (MSB)}, 5E \text{ (LSB)}$$

- Timer port address: LSB 24H & MSB 25H



Mode 1; M1=0, M2=1

– M1 M2 T13 T12 T11 T10 T8 T7

– 0 1 0 0 0 0 1 0 == (42H)

- Control word: (C0H)

D7 D6	D5	D4	D3 D2	D1	D0
Timer Command	IEB	IEA	PC	PB	PA

1 1	0	0	0 0	0	0
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Instructions to set counter & square wave generation



**MVI A , 5E ; LSB
of count**
**OUT 24H ; Load
LSB of timer
Register**
**MVI A, 42H ;
MSB count with**

Mode 1

MVI 25H ; Load MSB of timer Register

MVI A, C3H ; Load the control word for register

OUT 20H ; Trigger the counter by loading to Ctrl word to ctrl Reg

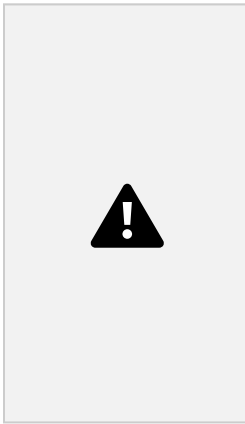
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Two Handshake signal (STB, IBF) • Steps in data input from IN device using 8155 – Peripheral put data in data line & send handshake signal STB

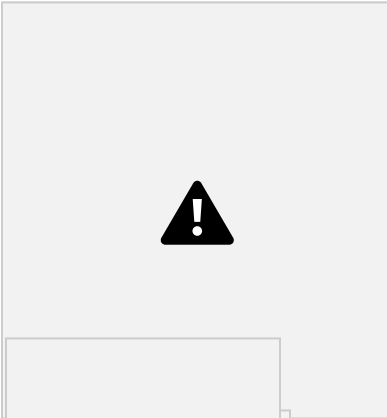
- Device inform Peripheral that IN port is full, don't send next byte until read by IBF signal
- Either MPU check status or Device interrupt to MPU for Reading data from Device





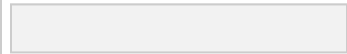
MPU

System Data Bus

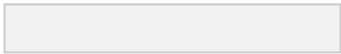


RD^b

Data lines

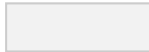


STB (Strobe)



IBF-IN Buff

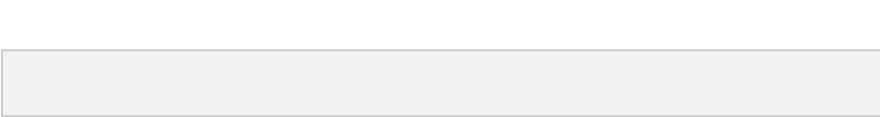
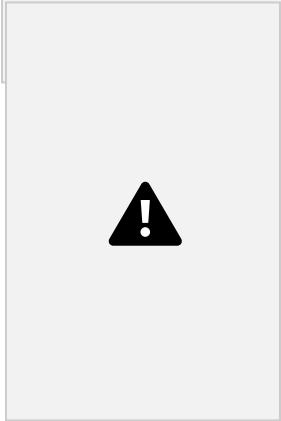
Full



Status

Peripheral (IN)

Such as Keyboard



Check

INTR



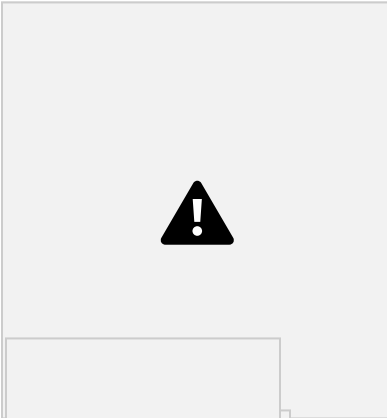
- MPU writes byte to the out port of Device by sending WR^b signal
- Device inform the peripheral by sending handshake OBF, that a byte on the way
- Peripheral ACK the byte by signal to device – Device Interrupt the MPU to ask to next byte or MPU check the status of Device





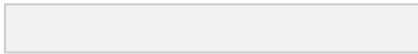
MPU

System Data Bus

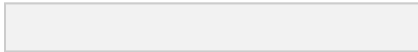


WR^b

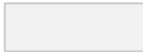
Data lines



OBF-OutBuff Full



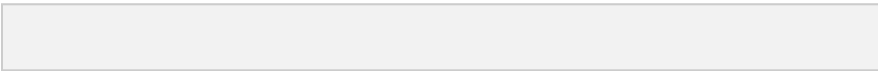
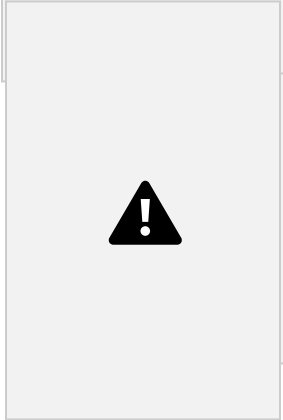
ACK



Status

Peripheral (OUT)

Such as Printer



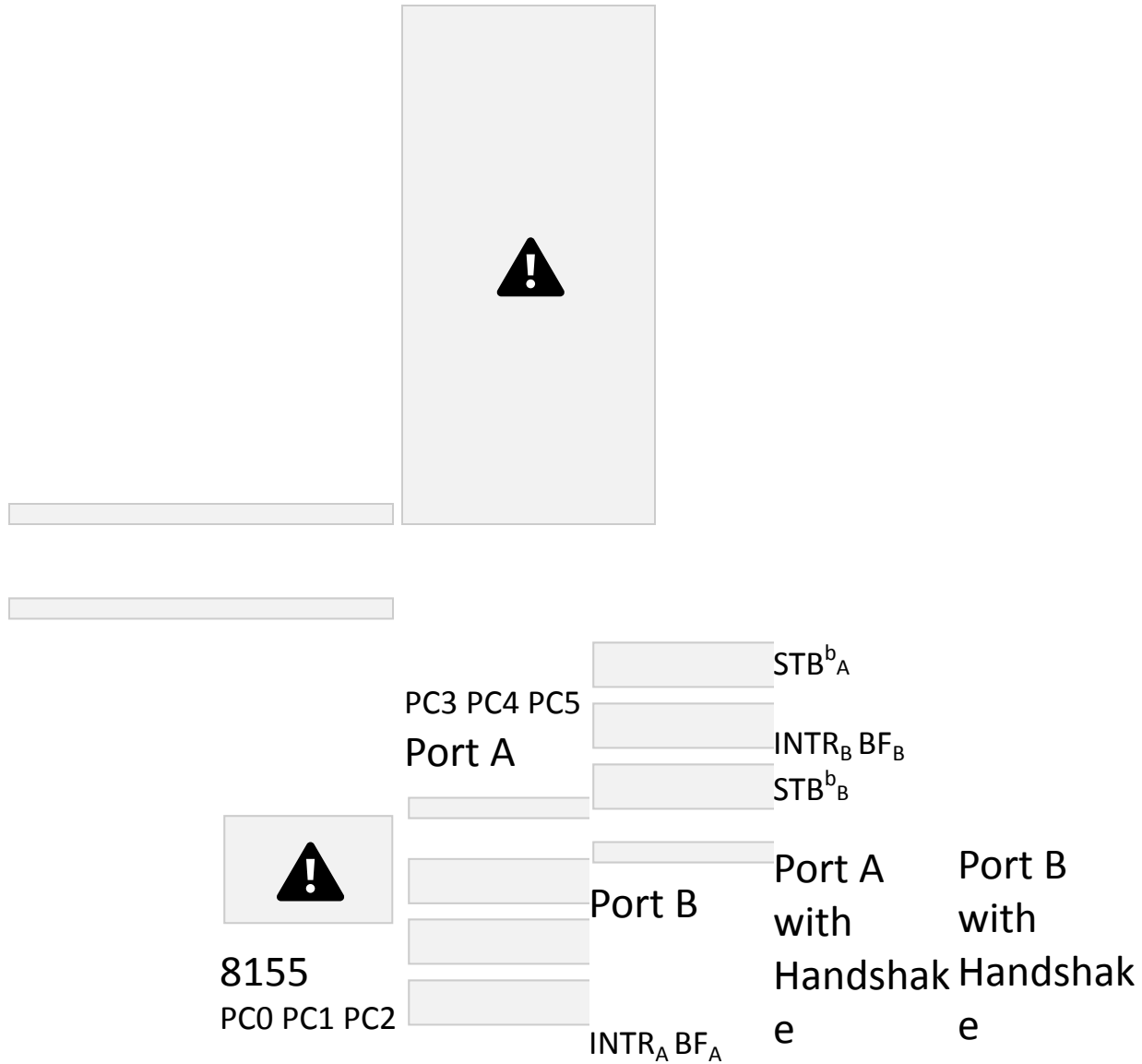
Check

INTR





- Port A and B: configured in Handshake mode
- Port A uses PC0, PC1, PC2 of Port C • Port B uses PC3, PC4, PC5 of Port C



- Control signals



STB^b(Strobe input):

- BF (Buffer Full):
- INTR (INTerrupt Request): Rising edge of STB^bif INTE =1
- INTE (INTerrupt Enable): D4 and D5 for Port A & B •

Input, Output : As discussed earlier

- Status word:

- MPU check the status Reg of port or timer
- Control register & Status register have same port
- Differentiated by RD^b and WR^bsignals



- Set up Port A in the handshake mode to read data from A/D Converter
- Setup port B as output port to display data at seven segment LEDs
- Use line PC3 from port C to initiate a conversion
- Use the 8155 Timer to record conversation time
- Control word

								CH=O/P CL=Handshake Mode			
D7	D6	D5	D4 D3	D2		D7	D6	D5	D4	D3	D2
0	0	0	0 0	1		No	X	X	X	X	X
Timer		NA (INTR not used)				Use	Timer	INTEb	BFb	INTRb	INTAa
		Use for Port C						Read the Data Mask with 02H			

- Status word

06H

- Timer (with effecting other I/O assignment)

– Start timer – **11000110010001 C6H 46H**

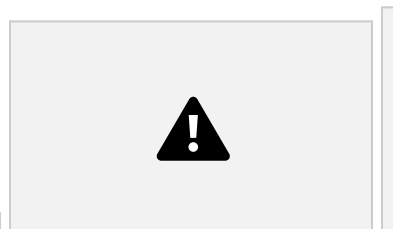
Stop timer **10**

A15 A14

A13 A12 A11



A2 A1 A0





3 to 8

Decoder

AD7 to

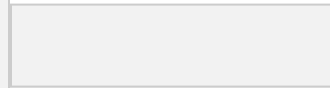
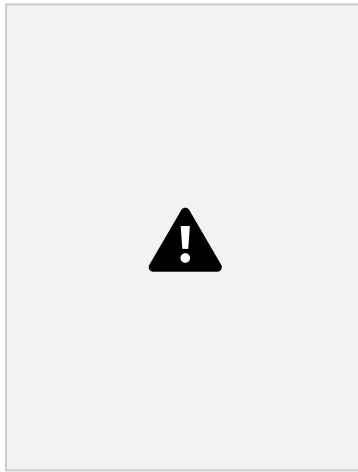
AD0 0₄



8155

PA7 PA6 PA5 PA4 PA3 PA2 PA1 PA0

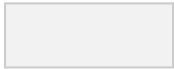
PC2 PC3



ADC
5790

DR^b B/C^b

V_i



Analog Input

PC3 is used for
Start
Conversion

IO/M^b

ALE

ALE

RD^b

WR^b

RESET OUT

WR^b

PB7

PB6

PB5

PB4

PB3

PB2

PB1

PB0

RESET OUT

IO/M^b

Port CL = HS
mode PC2 is
STROBE

Port CH = O/P

LED Driver

7
Seg
LED

7
Seg

MVI A,06H ;Control world for I/O port



ST:IN 20H ; Read Status Register ANI



MVI A,46H ; Byte to stop Counter
OUT 20H ;Stop Counter

**IN 21H ; Read A/D output OUT 22H
;Display data at port B (LEDs)**

INT 24H ; Read LSB of Timer

MOV L,A

INT 25H ; Read MSB of Timer ANI 3FH
;Mask the mode Bit D6,D7 MOV H,A ;
Save MSB timer count in H LHLD RWM ;
Store the count at ;Memory location
RWM HLT



- R S Gaonkar, “Microprocessor Architecture”, Chapter 14